

1) For infinitely many solutions, $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$

2) $4x + 6y - 9 = 0$ — (1)

$12x + my - 12 = 0$ — (2)

Given

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$$

$$a_1 = 4, \quad b_1 = 6, \quad c_1 = -9$$

$$a_2 = 12, \quad b_2 = m, \quad c_2 = -12$$

$$\therefore \frac{4}{12} = \frac{6}{m} \neq \frac{-9}{-12}$$

$$\frac{1}{3} = \frac{6}{m} \neq \frac{3}{4}$$

$$6 \times 3 = m$$

$$18 = m$$

$$\therefore \boxed{m = 18}$$

3) Intersecting lines.

$$4) \begin{array}{l} a_1 = 3 \\ a_2 = 6 \end{array} \quad \begin{array}{l} b_1 = 2 \\ b_2 = -4 \end{array} \quad \begin{array}{l} c_1 = -8 \\ c_2 = -9 \end{array}$$

$$\therefore \frac{a_1}{a_2} = \frac{3}{6} = \frac{1}{2}$$

$$\frac{b_1}{b_2} = \frac{2}{-4} = -\frac{1}{2}$$

$$\frac{c_1}{c_2} = \frac{-8}{-9} = \frac{8}{9}$$

$$\therefore \frac{a_1}{a_2} \neq \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$$

Hence these pair of equations are consistent.

$$5) \begin{array}{l} a_1 = 4 \\ a_2 = k \end{array} \quad \begin{array}{l} b_1 = 5 \\ b_2 = 10 \end{array} \quad \begin{array}{l} c_1 = 0 \\ c_2 = 0 \end{array}$$

Given,

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$$

$$\frac{4}{k} = \frac{5}{10} = \frac{0}{0}$$

$$\frac{4}{k} = \frac{1}{2}$$

$$8 = k$$

$$\boxed{k=8}$$

$$6) \quad a_1 = 2k, \quad b_1 = 5, \quad c_1 = -7 \\ a_2 = 6, \quad b_2 = -5, \quad c_2 = -11$$

Given,

$$\frac{a_1}{a_2} \neq \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$$

$$\frac{2k}{6} \neq \frac{5}{-5}$$

$$\frac{2k}{6} \neq -1$$

$$2k \neq -6$$

$$k \neq \frac{-6}{2}$$

$$\boxed{k \neq -3}$$

$\therefore$ k is any real number except -3 .

$$7) \quad 2x + 3y = 0$$

$$4x - 3y = 0$$

By elimination method,

$$2x + 3y = 0$$

$$4x - 3y = 0$$

$$6x = 0$$

$$\boxed{x = 0}$$

Substituting the value of x in eqn (1)

$$2(0) + 3y = 0$$

$$0+3y=0$$

$$3y=0$$

$$y=0$$

$$x+y$$

$$0+0=0$$

$$\begin{aligned} 2) -5x+7y &= 2 \\ -5x+7y-2 &= 0 \end{aligned}$$

$$a_1 = -5, b_1 = 7, c_1 = -2$$

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2} = \frac{1}{k}$$

$$\frac{-5}{a_2} = \frac{7}{b_2} = \frac{-2}{c_2} = \frac{1}{k}$$

$$\Rightarrow a_2 = -5k, b_2 = 7k, c_2 = -2k$$

Putting $k=2$,

$$a_2 = -5(2) = -10$$

$$b_2 = 7(2) = 14$$

$$c_2 = -2(2) = -4$$

$\therefore$ The second equation will be =

$$-10x+14y-4=0$$

$$-10x+14y=4$$

$$9) \quad 3x - 2y = 4 \quad \text{--- (1)}$$

$$3x - 2y \quad 6x + 4y = -8 \quad \text{--- (2)}$$

By elimination method,

$$(3x - 2y = 4) \times 6$$

$$(6x + 4y = -8) \times 3$$

$$18x - 12y = 24$$

$$18x + 12y = -24$$

$$18x = 0$$

$$\boxed{x = 0}$$

Substitute the value of x in eqn (1)

$$3(0) - 2y = 4$$

$$0 - 2y = 4$$

$$y = \frac{-4}{-2}$$

$$\boxed{y = -2}$$

$$\therefore x = 0, y = -2$$

$$10) \quad \frac{5}{x-1} + \frac{1}{y-2} = 2 \quad \text{--- (1)}$$

$$\frac{6}{x-1} - \frac{3}{y-2} = 1 \quad \text{--- (2)}$$

$$\text{Let } \frac{1}{x-1} = a \quad \text{--- (a)}$$

$$\frac{1}{y-2} = b \quad \text{--- (b)}$$

$$\therefore 5a + b = 2 \quad \text{--- (3)}$$

$$6a - 3b = 2 \quad \text{--- (4)}$$

By elimination method,

$$(5a + b = 2) \times 6$$

$$(6a - 3b = 2) \times 5$$

3

$$30a + 6b = 12$$

$$-(30a - 15b = 10)$$

$$21b = 2$$

$$\boxed{b = \frac{2}{21}}$$

Substitute the value of b in eqn (4)

$$6a - 3b = 2$$

$$6a - 3\left(\frac{2}{21}\right) = 2$$

$$\frac{6a}{1} - \frac{2}{7} = 2$$

$$\frac{42a - 2}{7} = 2$$

$$42a - 2 = 14$$

$$42a = 14 - 2$$

$$42a = 12$$

$$a = \frac{12}{42} = \frac{2}{7}$$

$$\boxed{a = \frac{2}{7}}$$

Substituting the value of a and b in eqn (a) and eqn (b) respectively

$$\frac{1}{x-1} = a$$

$$\frac{1}{x-1} = \frac{2}{7}$$

$$\boxed{x-1 = \frac{7}{2}}$$

$$2(x-1) = 7$$

$$2x - 2 = 7$$

$$2x = 9$$

$$\boxed{x = \frac{9}{2}}$$

$$\frac{1}{y-2} = b$$

$$\frac{1}{y-2} = \frac{2}{21}$$

$$\boxed{y-2 = \frac{21}{2}}$$

$$2(y-2) = 21$$

$$2y - 4 = 21$$

$$2y = 24 + 25$$

$$y = \frac{24}{2}$$

$$y = 12$$

$$\boxed{y = \frac{25}{2}}$$